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INVITED REVIEW

Benefits of a plant-based diet and considerations for the athlete

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Abstract
Individuals may opt to follow a plant-based diet for a variety of reasons, such as religious practices, health benefits or concerns for animal or environmental welfare. Such diets offer a broad spectrum of health benefits including aiding in the prevention and management of chronic diseases. In addition to health benefits, a plant-based diet may provide performance-enhancing effects for various types of exercise due to high carbohydrate levels and the high concentration of antioxidants and phytochemicals found in a plant-based diet. However, some plant-based foods also contain anti-nutrional factors, such as phytate and tannins, which decrease the bioavailability of key nutrients, such as iron, zinc, and protein. Thus, plant-based diets must be carefully planned to ensure adequate intake and absorption of energy and all essential nutrients. The current narrative review summarizes the current state of the research concerning the implications of a plant-based diet for health and exercise performance. It also outlines strategies to enhance the bioavailability of nutrients, sources of hard-to-get nutrients, and sport supplements that could interest plant-based athletes.

Keywords Vegan · Exercise · Physical activity · Vegetarian · Creatine · Supplements

Abbreviation
PB Plant-based

Introduction

Although a plant-based (PB) diet is not adequately defined in the literature in terms of nutrient and food content, it broadly refers to a variety of dietary patterns that emphasize higher intakes of plant foods and are low in animal products (McEvoy et al. 2012). These diets vary in their level of restrictiveness, from avoiding only red meat to the avoidance of animal products entirely (Clarys et al. 2014). The proportion of the population reporting following a PB diet continues to increase, with approximately 5–10% in developed countries reporting following some variety of PB diet (Nebl et al. 2019a). An overview of common PB diets is presented in Table 1. For the purposes of this review, a PB diet will refer to a diet void of all animal products (i.e., vegan) unless otherwise specified.

Rationales for following a PB diet are plentiful. Common motivations include religious practices, health benefits, ethical and moral concerns (solicitude around animal welfare and treatment), or environmental concerns (greenhouse gas emissions, less land, water, and energy usage compared to animal farming) (Mensink et al. 2016). Adoption of a well-planned PB diet in general or for specific clinical populations (cardiovascular disease, cancer, etc.) is nutritionally adequate and potentially superior to a traditional omnivorous diet for reducing the risk of chronic diseases, such as cardiovascular disease, type 2 diabetes, obesity, metabolic syndrome, hypertension, and some cancers (Kim et al. 2018, 2019; Tuso et al. 2013).

Despite the evidence for a PB diet's role in promoting good health, some doubt exists concerning a PB diet in

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Table 1 Classification of dietary patterns

Type of diet	Definition	Animal flesh	Fish	Dairy	Eggs
Non-vegetarian					
Omnivorous	Consumes all animal flesh and animal products (dairy, eggs, honey, gelatin, etc.)	✓	✓	✓	✓
Plant-based/Vegetarian					
Pescatarian	Consumes fish, dairy, and milk but not red meat, pork, or poultry	✗	✓	✓	✓
Lacto-Ovo vegetarian	Consumes eggs and dairy but no animal flesh	✗	✗	✓	✓
Lacto-vegetarian	Consumes dairy but no animal flesh or eggs	✗	✗	✓	✗
Ovo-vegetarian	Consumes eggs but no animal flesh or dairy	✗	✗	✗	✓
Vegan	Avoids all products derived from animals	✗	✗	✗	✗

**Adapted from Le & Sabaté, 2014*

supporting athletic training and performance (Lynch et al. 2018). While the inclusion of dairy into a PB diet generally ensures an adequate supply of nearly all nutrients required for optimal human health and performance, a diet void of other animal products could still be lacking micronutrients, such as calcium, vitamin B_{12}, and omega-3 fatty acids, which are crucial to not only health but also athletic performance (Kreider et al. 2010; Lukaski 2004). Therefore, proper dietary planning and consumption of a variety of PB foods are necessary to meet the nutritional requirements for all stages of life and fitness goals (Nebl et al. 2019a; Venderley and Campbell 2006). In this narrative review, we will summarize the current literature around the impact of a PB diet on health and performance and provide recommendations to optimize nutrition intake from a diet void of meat and animal products. Further, optimal training and performance will refer to the training and recovery supported by adequate nutrition to sustain and progress one's fitness and performance.

Impact of a plant-based diet for health and chronic disease

Like any diet, the nutrition quality of a PB diet lies on a spectrum from a minimally nutritious diet based on nutrient-poor, processed foods to a maximally nutritious diet rich in whole food sources. A well-planned, whole foods PB diet may decrease the incidence of chronic diseases, while a minimally nutritious PB diet rich in processed food may increase the risk of cardiovascular disease and diabetes (Satija et al. 2016, 2017). Therefore, a PB diet is no different from an omnivorous diet in that proper planning and attention must be given to the types of foods being consumed to ensure a nutritionally adequate diet that meets the individual's requirements for health, as well as fitness goals.

A systematic review by Schwingshackl et al. (2017) concluded that a diet rich in whole grains, vegetables, fruits, nuts, legumes, and fish and subsequently a reduced intake of red and processed meat (not the avoidance thereof) and sugar-sweetened beverages led to a decreased risk of premature death by ~ 80% when compared to intakes from the highest risk category. Similarly, Boeing et al. (2012) suggested that increased intakes of fruits and vegetables independently decrease the risk for various diseases due to polyphenolic and antioxidant properties, which influence inflammatory, cellular redox, endothelial, and metabolic processes, all of which are involved in the pathogenesis of various diseases. Similar findings were reported in an umbrella review by Angelino et al. (2019), who concluded that increased intakes of fruits and vegetables are likely to have protective effects against a variey of diseases with varying levels of evidence.

The high fiber content of a PB diet is also thought to have many beneficial effects on health through its impact on gut microbiota (Makki et al. 2018). Therefore, replacing a portion of one's animal-product intake with PB sources of food may enhance the quality of the diet by increasing the intake of polyphenols, antioxidants, some vitamins and minerals, and fiber while decreasing the intake of energy and less desirable fatty acids (Craddock et al. 2016).

Although few randomized controlled trials have investigated the effects of a strict PB diet, many epidemiological studies have determined the effect of replacing a portion of animal products with PB foods. Such practices have shown benefits for the prevention, morbidity, and mortality related to some chronic diseases (Alwarith et al. 2020; Ashaye et al. 2011; Cheung et al. 2014; Dauchet et al. 2006; Joshi et al. 2020; Sakkas et al. 2020; Qian et al. 2019; Yadav et al.

2016). However, much of the research for clinical conditions have involved the inclusion of more fruits, vegetables, and whole grains in the diet and not necessarily the avoidance of animal products. Further, an abundance of the evidence for a PB diet comes from epidemiological research and, therefore, a cause–effect relationship cannot be inferred.

Nutrient absorption in a plant-based diet

Although a PB diet is adequate for various ages and nutrition goals (Fuhrman and Ferreri 2010), a poorly planned diet may predispose an individual to deficiencies in energy and some nutrients due to decreased amounts or bioavailability in PB foods (Rogerson 2017). PB foods, such as legumes, grains, oilseeds, and nuts, may contain anti-nutritional compounds, such as phytates, tannins, and fiber. While mostly beneficial, such compounds decrease the absorption of some nutrients (Fig. 1; Shubham et al. 2020). Also, the protein quality of plant foods has to be considered.

Tannins

Tannin is a polyphenolic compound that is particularly astringent and found predominantly in grains, fruits, beans, pulses, nuts, and beverages, such as coffee, wine, and tea (Brglez Mojzer et al. 2016). Tannins create complexes with proteins, starch, and digestive enzymes in the gastrointestinal tract decreasing the absorption and utilization of nutrients, such as vitamins and minerals (Brglez Mojzer et al. 2016; Chung et al. 1998; Frutos et al. 2004; Shubham et al. 2020). However, despite decreasing the nutritional value of food, consumption of tannins also has anti-inflammatory, anti-oxidative, anti-microbial, and anti-carcinogenic effects on human health (Brglez Mojzer et al. 2016).

Phytate

Phytate is a mixed-cation salt of phytic acid (Myo-inositol 1,2,3,4,5,6-hexakis phosphoric acid) and acts as a storage form of phosphorus found in plant seeds such as field peas. In such seeds, ~ 60 to 80% of total phosphorus is stored as phytate (Warkentin et al. 2012). Phytate limits intestinal absorption of divalent cations by chelating minerals, such as calcium, magnesium, zinc, and iron, forming an insoluble complex that is ultimately excreted before the mineral can be absorbed (Bangar et al. 2017; Zhang et al. 2020). Despite the decrease in nutrient absorption, phytate also has health-promoting roles such as the prevention of diabetes, as an anti-carcinogenic agent, as an anti-oxidant, in the prevention of kidney stones, and the prevention of cardiovascular disease (Kumar et al. 2021).

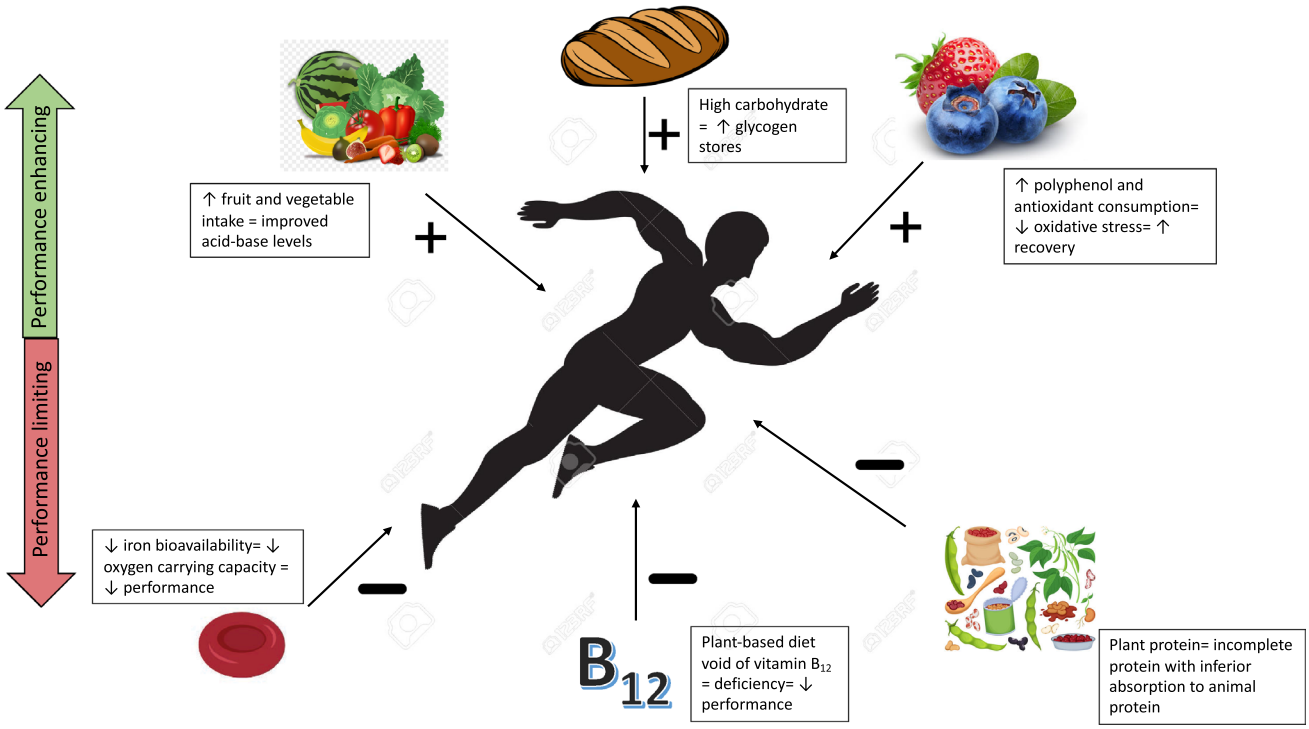

Fig. 1 Potential positive and negative implications of a plant-based diet on exercise performance

Fiber

Dietary fibers are a complex group of carbohydrates and lignin that cannot be digested by the small intestine. Dietary fibers have many beneficial impacts on human health, such as decreasing the risk of cardiovascular disease (Soliman 2019), enhancing the diversity of the gut microbiome (Makki et al. 2018), and regulating blood glucose (Vithana Pathirannehelage and Joye 2020). However, they also decrease nutrient absorption and increase satiety and, therefore, the athlete with high energy demands should be aware as to not overconsume fiber (Soliman 2019). Many physicochemical aspects of fiber, such as fermentation, bulking ability, binding ability, viscosity and gel formation, water-holding capacity, and solubility cause fiber to impact the absorption of some nutrients (Adams et al. 2018). Further, dietary fiber is known to slow digestion, increasing feelings of satiety and decreasing energy intake (He et al. 2022). While such charcteristics are beneficial for overall health (McRorie and McKeown 2017; Soliman 2019), it could make meeting the enerfy requirements of elite sport challenging (Melin et al. 2016). That said, if an athlete is looking to decrease body mass, fiber may be beneficial in helping them do so.

Protein quality

In general, the quality of a protein relates to the amino acids profile of a particular protein source and whether the source is able to provide sufficient supply of the indispensable amino acids (i.e., those that cannot be made in the body, or in sufficient amounts) to support protein turnover (Mariotti 2019). PB sources of protein are lower quality than animal-based proteins, as individual PB proteins are "incomplete", that is, they have reduced amounts of one of more of the nine indispensable amino acids whereas animal proteins are considered to be "complete" in that animal protein contains higher amounts of all these amino acids (Hoffman and Falvo 2004). For example, grains are deficient in lysine and threonine, pulses and legumes in methionine, tryptophan, and cystine, nuts and seeds in lysine, and tryptophan in maize (Hambræus 2014). As such, individuals consuming a PB diet are required to consume a variety of different types of protein sources to meet all of their indispensable amnio acid needs (Chardigny and Walrand 2016).

PB sources of protein also contain anti-nutritional factors, such as phytate and fiber, decreasing the digestibility of protein. For a detailed discussion on the impact of such anti-nutrional factors on protein digestibility, the interested reader is directed to Lynch et al. (2018). Some discrepancy exists within the literature as far as protein recommendations for PB diets. While some experts suggest PB individuals should consume 10% more energy from protein to make up for the 10% decreased bioavailability of PB sources (Kniskern and Johnston 2011), others have suggested that protein requirements are not dependant on the source of protein consumed (Agnoli et al. 2017).

Nutrition considerations for the athlete

Despite the evidence for the beneficial effects of a PB diet for general health and wellness, some continue to question its suitability for optimizing exercise performance, specifically around concerns of protein, iron, vitamin B_{12}, and creatine (Kaviani et al. 2020a). Conversely, others believe that a PB diet may be superior for exercise performance due to the high quantities of carbohydrates, phytochemicals, and antioxidants (Fig. 1; Craddock et al. 2016).

Energy

A well-planned, healthful PB diet is high in fiber and often includes high-volume, low-energy foods (Cialdella-Kam et al. 2016; Lis et al. 2019). As such, a PB diet is very satiating, leading to a decreased total energy intake. Decreased total energy intake is often attributed to a PB diet's beneficial health effects in some clinical populations (e.g., diabetes; Lynch et al. 2018; Newby 2009), but can be a concern for athletes with high energy requirements. If an athlete is not attentive to energy intake, they risk slipping into a state of low energy availability, leading to decreases in various measures of physical and mental health as well as performance (Mountjoy et al. 2018). Also, worth noting is that some athletes may adopt a PB diet to mask a restrictive eating disorder (Lis et al. 2019), so the integrated sport team (i.e., coaches, dietitians, and mental performance consultants) must be aware of the specific rationale for following a PB diet and monitor fluctuations in the athlete's weight. To ensure adequate energy is consumed, the athlete should monitor their weight and be aware of unnecessary or sizable degrees of weight loss. Some strategies for PB athletes to ensure adequate intake include eating five to eight meals and snacks per day, decreasing high fiber foods, and choosing energy-dense food choices, such as nuts and seeds, nut and seed butter, avocados, dried fruit, trail mix, hummus, and granola (Lis et al. 2019).

Protein

Although protein is found in abundance in PB sources (Table 2), the amino acid profile in plants is often suboptimal compared to animal-based sources (Lynch et al. 2018). In particular, PB sources of protein rarely match animal proteins for the branched-chain amino acids especially leucine. Leucine plays a key role in muscle protein synthesis through its role in stimulation of the mammalian target of rapamycin

Table 2 Plant-based sources with significant amounts of key nutrients. Adapted from Lis et al. (2019)

Compound	Plant-Based Sources
Protein	Beans, peas, lentils, chickpeas, soybeans, nuts
Omega-3 Fatty Acids[a]	Walnuts, chia, camelina, hemp, flaxseed, soybeans
Omega-6 Fatty Acids[a]	Corn, tortilla and potato chips, tofu, tempeh, nuts and seeds, vegetable oils, peanut butter
Monounsaturated fatty acids	Canola oil, olive oil, peanut oil, avocado, nuts
Iron	Beans, peas, lentils, edamame, chickpeas, nuts, seeds, whole grains, fortified bread and cereals
Zinc	Beans, peas, lentils, chickpeas, edamame, nuts, seeds
Calcium	Chinese cabbage, kale, texturized vegetable proteins, tofu, nuts, seeds, beans
Vitamin D	Fortified cereals, margarine, and plant-based "milk" (almond, soy, oat, pea, etc.)
Riboflavin	Nutritional yeast, quinoa, muesli, fortified cereals and plant-based "milk"(almond, soy, oat, pea, etc.), avocado, wild rice, and mushrooms
Vitamin B_{12}	Nutritional yeast, fortified plant-based "milk" (almond, soy, oat, pea, etc.), fortified meat analogs, shiitake mushrooms

[a]Polyunsaturated fat

(i.e., mTOR), a protein kinase that regulates protein synthesis and cellular growth (Devries et al. 2018; Hartman et al. 2007; Xu and Ren 2014). Research has observed decreased circulating branched-chain amino acids in both lacto-ovo vegetarians as well as vegans compared to their omnivorous peers (Wang et al. 2019), thus substantiating concerns regarding protein quality in PB protein sources. However, the Academy of Nutrition and Dietetics has reported in their 2016 position stand that consuming a wide variety of PB sources of protein and consuming adequate energy are likely sufficient to meet the required intakes of all indispensable amino acids (Melina et al. 2016).

When considering the literature investigating the impacts of PB proteins on muscle protein synthesis, the results are inconclusive. While Hevia-Larraín et al. (2021) concluded that both soy and animal protein are effective for gains in lean body mass and strength and that neither is superior, others have reported inferior muscle protein synthesis and lean mass accrual when consuming soy protein compared to milk protein post-exercise (Hartman et al. 2007; Wilkinson et al. 2007).

Despite these concerns, for both the athlete and non-athlete, widely accepted in the literature is that protein recommendations in developed countries can be met through consumption of a variety of PB sources, thus, sufficiently delivering all indispensable amino acids when caloric intake is adequate (Melina et al. 2016; Craig et al. 2009; Zello 2006). However, some athletes may find it challenging if to meet their protein needs from PB foods if their caloric requirements are exceptionally high. Due to the high fiber content, a PB diet tends to be very filling and, therefore, individuals may have a concern with consuming enough protein from whole food sources to support optimal muscle protein synthesis. If an athlete struggles to meet protein recommendations, a protein supplement may be a viable strategy. Multiple PB protein supplements have been studied and shown to be effective in enhancing hypertrophic response to exercise and/or exercise performance, such as soy (Candow et al. 2006), mung bean (Bartholomae et al. 2019), oat (Xia et al. 2018), rice (Joy et al. 2013), pea (Banaszek et al. 2019), and hemp (Kaviani et al. 2016b). A PB protein supplement formulated with a variety of protein sources may show benefit relative to supplementing with a single source to ensure all indispensable amino acids are consumed (Zello 2006). A supplement containing soy protein may be of particular interest due to its high leucine content (Rogerson 2017).

Older (e.g., Masters) athletes following a PB diet may want to pay particular attention to their protein intakes. The anabolic response to both resistance training and protein ingestion decreases with age, a phenomenon referred to as anabolic resistance. In brief, older adults experience decreased amino acid absorption and delivery following ingestion of protein, as well as impaired mTORC1 signaling in response to a bout of resistance training or protein ingestion (Burd et al. 2013). Despite decreased anabolic response in aging individuals, research has found that this can be overcome with a larger intake of protein. A meta-analysis of observational studies found that older adults that consumed very high (≥ 1.2 g/kg/day) and high (≥ 1.0 g/kg/day) protein intakes had better mobility as well as lower-limb functionality compared to those consuming low (< 0.8 g//kg/day), suggesting older adults may be better served with protein intakes greater than recommendations for younger individuals (Coelho-Júnior et al. 2018). Such recommended amounts may be challenging for older athletes consume if following a PB diet due to the high satiety of PB foods paired with a decreased appetite seen in older individuals (Strasser et al. 2021). However, a review by Berrazaga et al. (2019) suggested three main strategies to increase the anabolic

response of PB proteins that older athletes should consider if choosing to follow a PB diet: Increase protein intake to at least 30 g per meal, supplement with indispensable amino acids (leucine specifically), and blend PB protein sources to ensure high protein quality.

Carbohydrates

The high concentration of carbohydrates, and type of carbohydrate, found in a PB diet may offer performance advantages through assisting athletes in meeting carbohydrate recommendations for their activity levels, ensuring optimal glycogen concentrations, and preventing early fatigue (Barnard et al. 2019). Burke et al. (2017) observed not only improved race performance in trained race walkers but also improved exercise economy at a given pace in those consuming chronic high carbohydrate diets compared to those consuming a low high fat, low carbohydrate (i.e., "keto") diet. As such, the International Society for Sport Nutrition recommends athletes engaging in "moderate amounts of intense exercise" should consume 5–8 g/kg body weight carbohydrate daily to maintain liver and muscle glycogen (Kerksick et al. 2018). Given that muscle glycogen levels are directly correlated with time to fatigue in moderate (60–80% VO_2 max) intensity exercise, optimizing glycogen levels may delay fatigue in endurance and team sports (Hermansen et al. 1967; Kaviani et al. 2020b).

Healthful PB diets also tend to be rich in foods with a low glycemic index, which limits the increase of blood glucose following the ingestion of carbohydrate. A reduced blood glucose response would subsequently result in a decreased insulin concentration, facilitating lipolysis during exercise (Gao and Chilibeck 2020). Kaviani et al (2020b) observed an improvement in sport-specific performance as well as a decreased reliance on carbohydrate in recreational soccer players that consumed a low glycemic index sport bar made from lentils two hours before a simulated soccer game and at halftime compared to those that consumed a high glycemic index sport bar. However, others (Mizelman et al. 2020) observed no differences in distance covered during a soccer match in those consuming a pulse-based diet for four weeks compared to their regular (i.e., mixed) diet. Further, lentils, a low-glycemic index food, consumed before exercise tended to better preserve muscle glycogen during simulated soccer performance than a calorie- and macronutrient-matched meal containing potatoes and eggs (Little et al. 2010).

Athletes consuming a PB diet should be concious not to overconsume fiber, which is abundant in such a diet. Fiber intake has been correlated with gastrointestinal disturbances during competition, especially in endurance running (Jeukendrup 2017). High intakes of fiber may also impact protein digestibility. While the exact mechanisms by which fiber impacts digestibility is unclear, high fiber diets decrease protein digestibility in a linear manner (Adams et al. 2018).

Fatty acids

PB diets are typically lower in fat compared to the standard Western diet, a characteristic that has been attributed to many of its health benefits (Tuso et al. 2013). However, the type of fat and not the amount is more likely the source of the benefits found in following a PB diet. Mono- and polyunsaturated fatty acids play an important role in human health, especially in preventing and managing various diseases and inflammatory conditions and the absorption of some vitamins (Gómez Candela et al. 2011; Moreno and Salvadó 2000). A healthful PB diet is adequate in most omega-6 fatty acids (linolenic acid, arachidonic acid, gamma linoleic), while conjugated linoleic acid is less abundant. One must be aware not to overindulge in omega-6 fatty acids, however, as such dietary patterns have been linked to inflammation, vasconstriction, and platelet aggregation (Saini and Keum 2018). PB diets lack the omega-3 fatty acids eicosapentaenoic acid and docosahexaenoic acid most often found in fatty fish and contains limited sources of the omega-3 fatty acid α-linolenic acid, which can be found in PB sources such as flax (Table 2). Although α-linolenic acid can be converted endogenously to eicosapentaenoic acid and docosahexaenoic acid, it is a rather inefficient conversion and higly variable depending on factors, such as sex, dietary composition, health status, and age (Melina et al. 2016). Further, high intakes of linolenic acid may suppress conversion of α-linolenic acid (Plourde and Cunnane 2007; Rogerson 2017). To maximize conversion, intake ratios of 4:1 (linolenic acid to α-linolenic acid) are suggested (Melina et al. 2016). Philpott et al. (2019) reported supplementation with omega-3 fatty acids may be effective in improving muscle adaptation, energy metabolism, muscle recovery, and injury prevention regardless of the dietary patterns followed, so athletes looking to optimize their performance may want to pay close attention to ensure they are achieving optimal intake levels.

Micronutrients

Although all micronutrients except vitamin B_{12} can be found in adequate amounts in a PB diet, some must be paid particular attention. Supplementation may be desirable to prevent or treat nutrition deficiencies regardless of diet, such as iron, vitamin D, calcium, and vitamin B_{12}, and should be informed by a registered dietitian (Maughan et al. 2018; Larson-Meyer et al. 2018). For more detailed information on micronutrients often requiring supplementation in athletes, the interested reader is directed to Maughan et al. (2018).

Iron

Iron is an essential nutrient involved in oxygen transport, energy production, cell proliferation, and erythropoiesis (Sim et al. 2019). Although iron is found abundantly in a PB diet (Table 2), the bioavailability of PB sources (non-heme iron) is less than that found in animal sources (heme iron). Non-heme iron must be converted from its ferric form (Fe^{3+}) to a ferrous form (Fe^{2+}) before being absorbed, making it a more complex and less efficient physiological process compared to the absorption of heme iron, decreasing bioavailability (Fig. 2; Zimmermann and Hurrell 2007). Additionally, bioavailability may be decreased by the anti-nutritional factors discussed above (Sect. 3). Consequently, despite similar or greater iron intake, ferritin levels in PB individuals tend to be lower compared to omnivorous individuals.

Ferritin levels of PB individuals however typically fall within the normal range, potentially due to an adaptive response to a long-term PB diet that allows the individual to absorb non-heme iron more effectively (Melina et al. 2016). Further, some actions can enhance the body's absorption, including pairing non-heme iron with vitamin C or heme sources (if not entirely a PB diet) and avoiding foods rich in phytates, polyphenols, and calcium as well as exercise within two hours of consumption (Alaunyte et al. 2015; Zimmermann and Hurrell 2007).

Athletes tend to be at greater risk for iron deficiency due to increased inflammation from training inhibiting absorption, increased iron losses through sweat, and repeated foot contact on the ground damaging red blood cells (DellaValle and Haas 2012). Females may be at further risk due to iron losses through menstruation (Badenhorst et al. 2021). Further, increased iron intake may be required to support increased erythropoiesis (Pedlar et al. 2018). With increased iron needs in athletes and the high prevalence of iron deficiency across all populations, individuals may opt to supplement iron to maximize performance, especially endurance athletes (Nabhan et al. 2020; Sim et al. 2019). Athletes in a state of iron deficiency tend to have decreased aerobic power due to a reduction in oxygen-carrying capacity and, therefore, an increased reliance on anaerobic metabolism, with the degree of impairment positively correlated with the degree of deficiency (Sim et al. 2019). While those with iron deficiency anemia (iron deficiency with decreased hemoglobin mass) experience performance improvements when hemoglobin levels return to normal following iron supplementation (Pedlar et al. 2018), whether those with iron deficiency without anemia (iron deficiency with normal

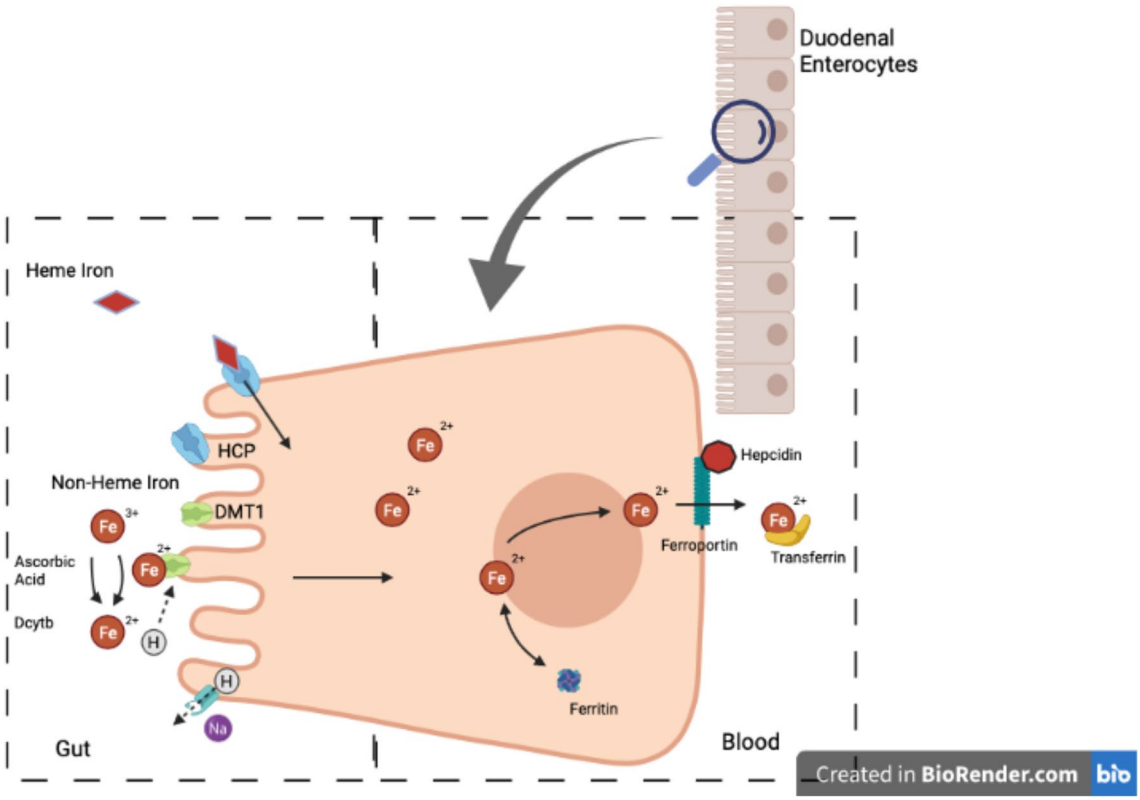

Fig. 2 Iron absorption in the duodenum. *HCP* Heme–iron transporter protein, *Dcytb* duodenal cytochrome B (reducing agent), *DMT1* divalent metal transporter 1, *H* hydrogen

hemoglobin mass) experience performance enhancements following supplementation remains equivocal, with the method of supplementation likely playing a role (Pedlar et al. 2018; Sim et al. 2019).

Zinc

Zinc is the second-most prevalent trace element in the human body after iron and plays an important role in various metabolic and immune processes (Hernández-Camacho et al. 2020; Saunders et al. 2013). Although zinc deficiency in developed countries is rare, PB individuals tend to have lower plasma zinc levels than omnivores (Melina et al. 2016) due to its decreased bioavailability from PB sources through the binding of phytate, which diminishes absorption. Despite this decreased bioavailability, PB individuals tend to have zinc levels within the normal range, possibly due to an adaptive homeostatic mechanism that enhances zinc absorption from PB sources (Saunders et al. 2013). Some food preparation techniques, such as soaking and sprouting beans, grains, nuts, and seeds and leavening bread, can decrease the binding of anti-nutritional factors to zinc, as well as iron, and increase bioavailability (Melina et al. 2016).

Athletes may want to pay particular attention to zinc bioavailability in their diet due to the role of zinc in energy metabolism, immunity, muscular contraction, and the lower levels of zinc typically observed in an active population compared to sedentary individuals (Chu et al. 2018; McClung, 2019). In particular, zinc plays an important role in regulating ion channels at the neuromuscular junction, leading to activation or inhibition of the corresponding ion currents, depending on zinc concentration and binding location (Hernández-Camacho et al. 2020). Individuals consuming a PB diet and active individuals may benefit from consuming more zinc to counteract decreased bioavailability and increased excretion, respectively (McClung, 2019), Limited research has investigated the use of zinc as an ergogenic aid, but that which is available is equivocal; therefore, aiding performance is in doubt (Heffernan et al. 2019). However, research has reported that individuals with insufficient dietary znc intake have decreased exercise performance in both strength and endurance (Van Loan et al. 1999) and aerobic power and exercise metabolism (Lukaski 2005). Taken together, depletion of zinc in the body may negatively impact exercise performance, but supplementing above the current recommended daily allowance (11 mg for men, 8 mg for women) is not warranted.

Calcium and vitamin D

Calcium is the most abundant mineral in the human body and is important for maximizing bone mineral accrual, blood clotting, nerve transmission, muscle contraction, and metabolism (Kreider et al. 2010). A common misconception is that individuals not consuming dairy products do not get adequate calcium, but a well-planned PB diet provides sufficient levels to meet the recommended daily allowance (Tuso et al. 2013). In addition to a variety of PB sources of calcium (Table 2), many PB foods are fortified with calcium to assist in meeting recommendations (Melina et al. 2016). However, the bioavailability of calcium from PB sources is often decreased due to high oxalate levels and, to a lesser degree, phytate. While consuming slightly higher levels of calcium may offset the decreased absorption of calcium from PB sources, simply consuming a variety of PB sources of calcium has been suggested to ensure adequate absorption (Hever, 2016).

Vitamin D is an imperative vitamin not only for bone health, immune function, and inflammatory modulation, but also is involved in muscular function and acts as an anabolic hormone (Antoniak and Greig 2017; Larson-Meyer and Willis 2010). Despite its crucial role in human health, wellness, and performance, 77% of Americans have insufficient levels of vitamin D, regardless of diet (Ginde et al. 2009). Athletes may be at risk for low levels of vitamin D if they participate in indoor sports, use excessive sun protection (i.e., sunscreen, clothing), if they live at high latitude, or if they have more skin pigmentation (Larson-Meyer and Willis 2010; Melina et al. 2016). Those following a PB diet may be at greater risk for inadequate vitamin D levels due to limited PB sources of vitamin D (Table 2), particularly if they do not achieve adequate sun exposure and/or have darker skin (Rogerson 2017). Those who receive sufficient sun exposure with adequate (but not excessive) sun protection may achieve optimal vitamin D levels through endogenously produced vitamin D in the skin. However, if maintaining adequate vitamin D levels is difficult and a synthetic supplement is not desired, PB vitamin D supplements derived from lichen, a composite fungal-algae organism, are commercially available (Rogerson 2017).

B vitamins

Although all of the B vitamins are crucial for optimal health and performance, only vitamins B_2 and B_{12} will be discussed here due to their scarcity in a PB diet. Riboflavin (vitamin B_2) is an important nutrient involved in energy metabolism and participates in redox reactions in various metabolic pathways with derivatives of the nutrient acting as electron carriers (Peechakara and Gupta 2021). Given its role in ATP generation, a riboflavin deficiency is likely to cause a disturbance at the intermediate steps of metabolism and, therefore, impact energy availability during aerobic exercise (Kreider et al. 2010). Riboflavin is found predominantly in organ meats and eggs, however, PB sources do provide amounts adequate in meeting the recommended daily

allowance (Table 2). Also, Vit B_2 is often found fortified in PB foods (e.g., cereal, bread). Exercise may increase the requirements for riboflavin intake, but the additional intake needed to cover increased needs is small and easily achievable through nutritious food choices (Manore 2000). Despite riboflavin's key role in various metabolic processes, research is inconclusive on whether supplementation with riboflavin has the potential to improve performance and any current research available varies in quality (Gonçalves and Portari 2021).

Vitamin B_{12} (cobalamin) is an essential nutrient necessary for normal nervous system function, homocysteine metabolism, and DNA synthesis and is important for red blood cell formation and transmission of neural signals, aspects that may be of particular interest for athletes (Krzywański et al. 2020). Vitamin B_{12} is found solely in animal products and, therefore, requires supplementation. Marginal amounts of vitamin B_{12} may be available through nutritional yeast and fortified cereals and PB "milk" (e.g., soy, almond, pea, oat), but such sources will not provide sufficient intake (Rogerson 2017). In addition to detrimental health effects, vitamin B_{12} deficiency is associated with impaired methionine production, leading to lower creatine biosynthesis and potentially impaired physical performance (Mahmood 2014).

Creatine

Creatine is a component of phosphocreatine, which is an important fuel source during high-intensity, short-duration exercise. In an omnivorous diet, approximately 1 g per day of creatine is consumed through meat and fish, and an additional 1 g per day is synthesized endogenously through arginine, glycine, and methionine metabolism, which appears to be sufficient to maintain creatine stores (Kaviani et al. 2020a; Rogerson 2017). However, creatine is limited in a lacto-ovo vegetarian diet and devoid in a strict PB diet, leading to decreased creatine concentrations in serum, plasma, red blood cells, and muscle, but not in the brain of PB individuals, which may result in suboptimal fuel available for alactic anaerobic metabolism and subsequent decreases in performance (Solis et al. 2015; Wallimann et al. 2011). Therefore, supplementation of creatine might be considered to optimize short-duration, high-intensity exercise performance in those following a PB diet. PB individuals can increase their phosphocreatine levels with creatine supplementation to a greater extent than omnivores due to a "super-compensation" effect, despite no differences in the muscle creatine transporter (Watt et al. 2004). Some studies indicate that vegetarians can enhance their physical performance (i.e., work output during a repeated-contraction isokinetic test) compared to omnivores following creatine supplementation (Burke et al. 2003), while others show no greater benefit for enhancement in exercise performance in vegetarians compared to omnivores when supplementing with creatine (Watt et al. 2004). Although a minimum of 1 g of supplemental creatine per day is necessary to match creatine intakes consumed in the average omnivorous diet (Kaviani et al. 2020a), supplementation with 5 g/day is associated with optimal tissue saturation (Maughan et al. 2018).

In addition to providing fuel for short duration, high-intensity exercise, creatine supplementation may also have beneficial effects on cognitive function and concussion recovery (Roschel et al. 2021). As such, athletes that participate in team skill-based sports may benefit from creatine supplementation. Some have suggested that vegetarians and vegans may experience greater enhancements in cognition following creatine supplementation compared to their omnivorous peers (Benton and Donohoe 2011; Rae et al. 2003), however, due to methodological concerns, more research is required. Athletes in activities at increased risk for concussions may also seek to maximize their creatine stores through supplementation as a precautionary strategy to improve their recovery should a concussion be sustained (Antonio et al. 2021).

When considering a PB diet in older athletes, some research has reported lower muscle creatine content in older individuals compared to a younger cohort (Forsberg et al. 1991; Möller et al. 1980), potentially due to a decreased number/size of type II muscle fibers, types and intensity of exercise, and differences in dietary intake (Gualano et al. 2016; Chilibeck et al. 2017). However, others have observed no differences with aging or an increase in creatine content in older adults (Kent-Braun and Ng 2000; Rawson et al. 2002). A recent meta-analysis by Forbes et al. (2021) reported significant increases in lean tissue as well as both upper and lower body strength indices in older adults consuming creatine with resistance training. Further, creatine has been hypothesized to play a role in the maintenance of bone health and the reduction of fractures with increasing age (Candow et al. 2019; Heyden and From 1970). However, a meta-analysis by Forbes et al. (2018) reporting no significant improvement in bone at any of the sites measured (lumbar spine, femoral neck, and whole-body).

Beta-alanine

Beta-alanine is the rate-limiting precursor to carnosine, an intracellular buffering agent (Derave et al. 2007; Peeling et al. 2018). Beta-alanine is found in meat and poultry, thus not consumed by those avoiding such products. Although beta-alanine can be created endogenously, individuals consuming PB diets have lower muscle carnosine concentrations than omnivores due to decreased intakes of beta-alanine. Carnosine is an important molecule for exercise performance, as it is an intracellular buffer, acting as the immediate defense against proton accumulation in the contracting

musculature during exercise, delaying fatigue (Maughan et al. 2018). Therefore, decreased levels may negatively impact performance (Rogerson 2017). Maughan et al. (2018) showed that supplementing with ~ 65 mg beta-alanine per kilogram over multiple doses per day for 10–12 weeks had a small but meaningful benefit in exercise lasting 30 s–10 min.

Taurine

Taurine is a conditionally indispensable amino acid concentrated in skeletal muscle and plays a role in osmoregulation and neurotransmission (Yatabe et al. 2009). Taurine is produced endogenously in the liver but can also be consumed through intakes of meat, seafood, and dairy products. PB individuals may consume negligible amounts, leading to potentially low or deficient levels (Fuhrman and Ferreri 2010; Rogerson 2017). Taurine is thought to play a role in exercise metabolism, as decreased taurine concentrations have been observed in rat muscle following exhaustive exercise (Yatabe et al. 2009). Taurine has a role in glucose tolerance, insulin sensitivity, and substrate uptake, storage, and oxidation in skeletal and cardiac muscle, all of which may impact exercise performance (Galloway et al. 2008). In humans, Rutherford et al. (2010) found that supplementation with 1.66 g of taurine one hour prior to a bout of cycling had no effect on time trial performance but did increase fat oxidation. da Silva et al. (2014) found that ingestion of 50 mg/kg/day of taurine for 14 days before a bout of eccentric exercise and for seven days after resulted in improved performance and decreased markers of muscle damage and oxidative stress. Such physiological effects have led some researchers to suggest that PB athletes should supplement with 500 mg taurine twice daily to maximize performance (Fuhrman and Ferreri 2010), although this type of supplementation protocol has not been reported in the literature.

Impact of plant-based diets on exercise performance

Several hypotheses have been presented to suggest a PB diet may be either superior or inferior to a meat-containing diet for exercise performance. While some have suggested the high prevalence of carbohydrates (Sect. 4.5) and polyphenols may lead a PB diet to be superior for performance than one containing animal products, others have questioned the appropriateness of such eating patterns, with particular concern around protein quality and creatine, particularly when considering muscle protein synthesis (Fig. 1). While these concerns have been addressed above, the following sections will examine the factors thought to enhance performance as well as the literature on performance outcomes in those following a PB versus omnivorous diet.

Antioxidants and polyphenols

A PB diet may enhance endurance exercise performance through the high levels of polyphenols and antioxidants, compounds that may help reduce the amount of oxidative stress and inflammation accrued during exercise, enhancing recovery and overall health and wellness (Bowtell and Kelly 2019). Further, such compounds may play a role in augmenting vessel diameter and enhancing blood flow, thus enhancing the delivery of oxygen and nutrients to working muscles while removing metabolic by-products (Potì et al. 2019; Roelofs et al. 2017).

A paucity of high-quality research makes drawing firm conclusions difficult (Higgins et al. 2020). A review by Merry and Ristow (2016) reported no evidence to support the use of supplemental antioxidants for exercise performance. While research supports the use of supplemental antioxidants improving performance in those with known deficiencies (Paschalis et al. 2016, 2018), a growing body of literature suggests that such supplementation may blunt the adaptations to exercise through reductions in mitochondrial biosynthesis, insulin sensitivity, and muscular hypertrophy in those with no known deficiency. However, most of the research in this sphere utilizes supplemental antioxidants, not whole food products alone.

Body composition

An abundance of evidence exists in support of a PB diet for leaner body composition and decreased body mass (Chen et al. 2019; Yadav et al. 2016; Sabaté and Wien 2010), characteristics that athletes may strive for, especially if they compete in weight class or esthetic sports or sports where weight has a direct impact on performance (i.e., cycling). Phillips et al. (2004) found that adherence to a PB diet decreased skinfold thickness and waist-to-height ratio, independent of weight loss. Further, consuming a post-exercise meal consisting of lentils has favorable effects on fat oxygenation compared to a post-exercise meal consisting of potatoes and eggs; this was attributable to the low glycemic index of lentils and other PB sources of protein (Kaviani et al. 2016a). This enhanced fat oxidation, over time, may contribute to improved body composition.

A PB diet may be beneficial for athletes interested in decreasing their body mass and/or adiposity through the consumption of nutrient-rich foods with low energy density, typical of a healthy PB diet. Such characteristics help with weight management through increased satiation at a lower caloric intake compared to foods with a lower volume to energy ratio (Boeing et al. 2012). Additionally, a PB diet's high fiber content may help decrease energy intake by slowing gastric emptying and energy/nutrient absorption,

increasing satiety, and influencing fat oxidation and storage (Slavin 2005).

When considering the accrual of lean tissue, the evidence is much more equivocal. While a meta-analysis by Messina et al. (2018) reported no differences between soy and animal protein supplementation for strength or lean mass gains in response to resistance training, others have reported superior muscle protein synthesis and lean tissue mass accretion with milk-based proteins compared to soy (Hartman et al. 2007; Wilkinson et al. 2007). Hevia-Larraín et al. (2021) reported no differences in strength or muscle mass accrual in those consuming a PB diet and supplementing with a soy protein isolate compared to a protein-matched omnivorous diet supplementing with whey protein. Further, some have suggested increasing amount of protein conxumed at each meal as well as throughout the day when using PB protein and mixing specific/complementary sources of PB protein to create a more balanced amino acid profile from PB sources (Berrazaga et al. 2019; Pinckaers et al. (2021).

Impact on exercise performance

Many of the concerns around the adequacy of a PB diet for athletes relate to performance in strength or power-based sports, in particular protein quality and quantity, as well as creatine intake. However, limited evidence is available characterizing the impact of a PB diet such types of exercise performance. If intake of key nutrients required for the performance or recovery of strength and power exercise, such as creatine, carnosine, and protein, is inadequate, performance and recovery may be limited (Lynch et al. 2018). As discussed above, although these are real concerns, adequate consumption of all three of these may be obtained either through dietary, including PB, or supplemental sources. The limited evidence available suggests that PB diets may produce similar gains in strength compared to omnivorous diets when matched for protein (Hevia-Larraín et al. 2021; Messina et al. 2018).

Many of the potential benefits of a PB diets, such as improved blood flow, reduced oxidative stress and inflammation, and improved glycogen storage, may directly impact endurance exercise performance (Barnard et al. 2019). However, despite the wide variety of potential mechanisms that may support a PB diet as beneficial for exercise performance, current literature suggests that neither a PB nor an omnivorous diet is superior for aerobic exercise performance (Craddock et al. 2016; Nebl et al. 2019b; Venderley and Campbell 2006). While some research suggests PB athletes may have higher aerobic power (VO_2 max) scores compared to omnivores (Boutros et al. 2020), the same findings do not hold when nutrient intake, training status, body weight, and other confounders are controlled for (Nieman 1999; Venderly and Campbell 2006). Further, PB diets tend to be more alkaline, due to higher intakes of fruits and vegetables. However, dietary manipulations do not significantly effect cellular and tissue pH as it is tightly regulated and any changes pH are minimal and controlled for (Applegate et al. 2017; Robergs et al. 2018) Limited evidence is available characterizing the impact of a PB diet on anaerobic, strength, or power performance; however, if the above-noted considerations are taken into account and supplementation occurs as indicated, there is no reason to suggest that such exercise should be impaired by a PB diet.

Conclusion

This review summarized the role of a PB diet in general health and exercise performance and outlined specific nutritional considerations for the athlete following a PB diet. Although a PB diet has favorable health effects in both clinical and healthy populations, such benefits are likely related to an increased intake of fruits, vegetables, and whole grains rather than the avoidance of animal products. However, if not planned properly, a PB diet may provide insufficient amounts of certain macro- and micro-nutrients, and thus care should be taken when choosing to follow a PB diet to ensure adequate intake and absorption of hard-to-get nutrients. Further, in athletes, a healthful PB diet (rich in whole food sources and limited in processed foods) neither hinders nor aids exercise performance if properly planned and implemented, although some actions can be taken to optimize performance (e.g., supplementation). An athlete following a PB diet should work closely with a registered dietitian with a background in sport to ensure optimal energy and nutrient intake and absorption.

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